

The resting membrane potential

Name _____ Date _____

Use this sheet while you work through the resting membrane potential walkthrough. It has 28 predictions.

1. When a step asks you to predict, write or draw your prediction in the top box, in pencil or your first color. Type it into the worksheet beside the walkthrough too. Next stays locked until you press Show me.
2. Press Show me. In the bottom box, in a second color, write what actually happened and whether you were right. If you were not, write what you would change.
3. When you finish the walkthrough, close it and do Part 2 from memory. Then reopen it and correct Part 2 in your second color. What you had to fix is what to study.

Part 1. Your predictions

PREDICTION 1 OF 28 · START WITH A BALLOON

If these balls could pass through the balloon, which way would they tend to move?

My prediction, before Show me

What happened, and was I right?

PREDICTION 2 OF 28 · THE MEMBRANE BLOCKS THEM

If the membrane stops the balls, what would they need in order to cross?

My prediction, before Show me

What happened, and was I right?

PREDICTION 3 OF 28 · A DOOR FOR POTASSIUM

The door is open. What will the potassium do?

My prediction, before Show me

What happened, and was I right?

PREDICTION 4 OF 28 · EVERY BALL CARRIES A CHARGE

Potassium is leaving and taking its positive charge with it. The negative proteins stay. What happens to the charge inside the cell?

My prediction, before Show me

What happened, and was I right?

PREDICTION 5 OF 28 · A TUG OF WAR AT THE DOOR

What are the two pulls, which way does each one go, and will all of the potassium eventually leave?

My prediction, before Show me

What happened, and was I right?

PREDICTION 6 OF 28 · MEET SODIUM

Look at a sodium ball outside the cell. Which way does the crowding push it, and which way does the negative inside pull it?

My prediction, before Show me

What happened, and was I right?

PREDICTION 7 OF 28 · ONLY A FEW SODIUM DOORS

A little sodium trickles in through the sodium door. What does that do to the charge inside the cell?

My prediction, before Show me

What happened, and was I right?

PREDICTION 8 OF 28 · POTASSIUM HAS THE MOST SAY

A drug blocks every potassium door. Does the inside stay just as negative?

My prediction, before Show me

What happened, and was I right?

PREDICTION 9 OF 28 · THE LEAKS NEVER STOP

If nothing stopped the leaks, what would happen to the crowding on each side over many hours?

My prediction, before Show me

What happened, and was I right?

PREDICTION 10 OF 28 · TELL THE STORY SO FAR

Tell the whole story so far: why does the inside of a resting cell end up negative, and what keeps it that way?

My prediction, before Show me

What happened, and was I right?

PREDICTION 11 OF 28 · MOST DOORS HAVE GATES

Think about what a cell might need to respond to. What kinds of things could open a door?

My prediction, before Show me

What happened, and was I right?

PREDICTION 12 OF 28 · THE LINE STARTS TO RISE

As the line rises toward zero, is the inside becoming more negative or less negative?

My prediction, before Show me

What happened, and was I right?

PREDICTION 13 OF 28 · THE WORDS DESCRIBE THE CHANGE

At the peak the inside is positive, and then it starts moving back toward zero while it is still positive. Which word describes that change?

My prediction, before Show me

What happened, and was I right?

PREDICTION 14 OF 28 · HOW MUCH OF EACH ION

Which ion is the odd one out in this table, and what else inside the cell carries charge but can never leave?

My prediction, before Show me

What happened, and was I right?

PREDICTION 15 OF 28 · TWO PUSHES, ONE TOTAL

The inside is negative. For potassium and for sodium, which way does the crowding push, and which way does the charge pull?

My prediction, before Show me

What happened, and was I right?

PREDICTION 16 OF 28 · POTASSIUM'S BALANCE POINT

The resting cell is at -70 mV . If potassium were the only ion that could cross, would its balance point be more negative or less negative than -70 ?

My prediction, before Show me

What happened, and was I right?

PREDICTION 17 OF 28 · READING THE EQUATION

Outside potassium rises from 5 to 10 mM. Does potassium's balance point get more negative or less negative?

My prediction, before Show me

What happened, and was I right?

PREDICTION 18 OF 28 · THE RULE

The membrane is at -70 mV and sodium doors open. Which way does sodium move, and which way does the membrane go?

My prediction, before Show me

What happened, and was I right?

PREDICTION 19 OF 28 · SAME RULE, NEW STARTING POINT

At $+30\text{ mV}$, potassium doors open. Which way does potassium move, and how hard?

My prediction, before Show me

What happened, and was I right?

PREDICTION 20 OF 28 · THE GOLDMAN EQUATION

A drug opens extra potassium leak doors. Where does the membrane go?

My prediction, before Show me

What happened, and was I right?

PREDICTION 21 OF 28 · OPEN THE SODIUM DOORS INSTEAD

Many sodium doors open at once. Where does the membrane go?

My prediction, before Show me

What happened, and was I right?

PREDICTION 22 OF 28 · WHAT THE PUMP DOES, AND WHAT IT DOES NOT

A poison stops the pump completely. Does the membrane potential collapse right away? What happens if you wait several hours?

My prediction, before Show me

What happened, and was I right?

PREDICTION 23 OF 28 · REST IS THE STARTING LINE

Three neurons share a threshold of -55 mV. Cell A rests at -70 , cell B at -80 and cell C at -62 . Rank them from easiest to hardest to fire.

My prediction, before Show me

What happened, and was I right?

PREDICTION 24 OF 28 · PATIENT 1: MISSED DIALYSIS

His kidneys are not removing potassium, so his blood potassium is 7.0 mM (normal 3.5 to 5.0). What happens to potassium's balance point, and then to his resting membrane potential?

My prediction, before Show me

What happened, and was I right?

PREDICTION 25 OF 28 · PATIENT 1: THE TREATMENTS

For each treatment, decide whether it protects the heart, lowers the potassium in his blood, or removes potassium from his body.

My prediction, before Show me

What happened, and was I right?

PREDICTION 26 OF 28 · PATIENT 2: THE WATER PILL

Her blood potassium is 2.8 mM (normal 3.5 to 5.0). Predict what happens to potassium's balance point, to her resting membrane potential and to how easily her muscles fire.

My prediction, before Show me

What happened, and was I right?

PREDICTION 27 OF 28 · PATIENT 3: THE NUMBED SKIN

Does lidocaine change the resting membrane potential of the pain nerves in that skin? Why does the patient feel no pain from the needle?

My prediction, before Show me

What happened, and was I right?

PREDICTION 28 OF 28 · THE WHOLE STORY, WITH THE REAL NAMES

Explain the resting membrane potential from start to finish: where the ions are, why the inside is negative, why rest is about -70 mV , and what the pump does.

My prediction, before Show me

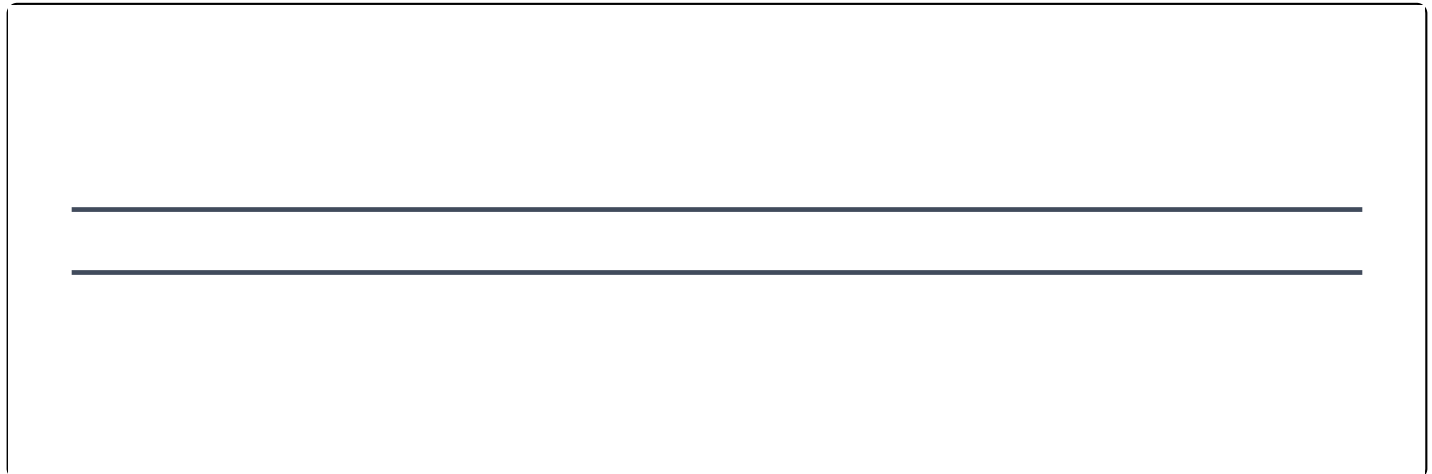
What happened, and was I right?

Part 2. Draw it from memory

The resting membrane potential

1. Set up the cell

The membrane is drawn for you. Put potassium, sodium, and the large negative proteins where they actually sit in a resting cell, using more symbols where there is more of that ion. Draw in the potassium leak channels and the few sodium channels. Mark the inside and the outside.



Write the approximate concentration of potassium and sodium on each side, with units.



2. Draw the two forces on potassium

Pick one potassium ion sitting in a leak channel. Draw an arrow for the chemical force on it and a different arrow for the electrical force. Label which is which, and mark which arrow is winning at rest.



3. Explain the number

A resting neuron reads about negative seventy millivolts. In your own words, say why it is negative, why it is closer to potassium's balance point than to sodium's, and what the pump is and is not doing.